Lesson 9: Coupling

ChatGPT edition — send a pulse without a wire between coils

Geometry controls shared changing flux

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

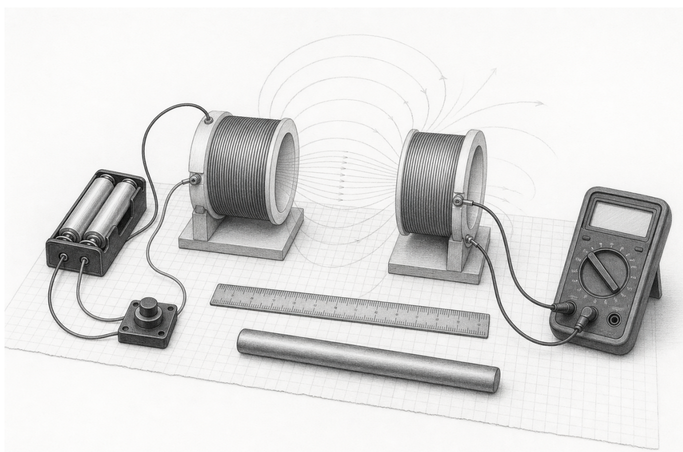
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Two insulated coils
- 2 AA cells and pushbutton
- Multimeter
- Iron core
- Ruler

Predict first

Which matters more to the received pulse: coil separation or wire length between the meter and receiver?



Change one geometric variable at a time: gap, orientation, or the removable common core. Faint contours show that only part of the transmitter's changing flux links the receiver.

Investigate

1. Drive the first coil only with brief make/break pulses. Measure the second coil's response at fixed alignment.
2. Increase separation in measured steps.
3. Repeat at 90-degree orientation.
4. Insert the common iron core and repeat the distance series.

Explain from the evidence

Mutual induction depends on how much changing flux from the transmitter links the receiver. Separation, orientation, turns, core, and drive-change rate all matter. A weak pulse at poor alignment is a measurement, not failed magic. Energy still comes from the battery; coupling controls how much reaches the second circuit.

Change one thing

Create a two-dimensional coupling map by moving the receiver over graph paper while keeping pulse method fixed.

Evidence record	
Prediction	______________________________
One change	______________________________
Observation	______________________________
Claim + because	______________________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.